

TUTORIAL 5 (FP2)
CHAPTER 5: DIFFERENTIAL EQUATIONS
Section I – Notes and Exercises

5.0: RECAP OF THE BASICS

Differential Equations

1. A differential equation in x and y is an equation expressing the relationship between an **independent variable** x , a **dependent variable** y and the **derivatives** of y with respect to x .
2. The following are some examples of differential equations:
 - $y \frac{dy}{dx} + x^2 = 0$
 - $xy \frac{d^2y}{dx^2} - \sin x = 0$
 - $x \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} = y$

Order of a Differential Equation

1. The order of a differential equation is the **order of the highest derivative** of the unknown function involved in the equation.
2. Examples of **first order** differential equations:
 - $\frac{dy}{dx} + 2x = y$
 - $y \frac{dy}{dx} - \cos x = 0$
3. Examples of **second order** differential equations:
 - $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + y = 0$
 - $e^x \frac{d^2y}{dx^2} = 2y$

General Solution and Particular Solution of a Differential Equation

1. Given a differential equation $\frac{dy}{dx} = 3x^2$, the corresponding solution $y = x^3 + C$, where C is an **arbitrary constant**, is known as **general solution** of the given differential equation.
2. Now, suppose that we are given a differential equation $\frac{dy}{dx} = 3x^2$ and the **condition** when $x = 0, y = 1$.
 Substituting these values of x and y into its general solution $y = x^3 + C$, we get $C = 1$.
 Hence, $y = x^3 + 1$ is the **particular solution** of the differential equation.

5.2: SECOND ORDER DIFFERENTIAL EQUATIONS

Notation and Vocabulary

- As an example, the differential equation

$$\frac{d^3y}{dx^3} - 7\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = 3\sin x$$

is described as

- **third order** (because the highest derivative is a third derivative)
 - **linear** (because where y and its derivatives appear, they are to the power 1)
 - having **constant coefficients** (because the coefficients of the terms involving y are constants – in this case 1, -7 , 2 and 4)
 - **non-homogeneous** (because the **right-hand side**, the part not containing y , is **not zero**). In cases where the right-hand side **is zero**, the differential equation is called **homogeneous**.
- In this chapter, you will meet **second order, linear differential equations, with constant coefficients**. In general, they can be written as follows:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

where a , b and c are constants.

The Auxiliary Equation Method

- Before solving second order differential equations, you will find it helpful to look at the form of first order, linear, homogeneous differential equations with constant coefficients. In general, they are of the form:

$$\frac{dy}{dx} + ky = 0$$

You can solve this by separating the variables:

$$\frac{dy}{y} = -ky$$

$$\left(\frac{1}{y}\right) \frac{dy}{dx} = -k$$

$$\int \frac{1}{y} dy = -k \int 1 dx$$

$$\ln |y| = -kx + c$$

$$y = \pm e^{-kx+c}$$

$$= \pm e^c e^{-kx}$$

$$= A e^{-kx} \text{ where } A = \pm e^c$$

- So, in general, the solution of any first order equation of this type will contain **an exponential function** and **one unknown (arbitrary) constant**.
- Knowing the form of the solution allows you to find it without doing any integration, as shown in the next example.

2nd Order RLC circuit

- Circuit w at least 1 capacitor & 1 inductor

Imaginary Number Revision

Cartesian form $z = x + jy$ → Not "i" cuz that edy used for current

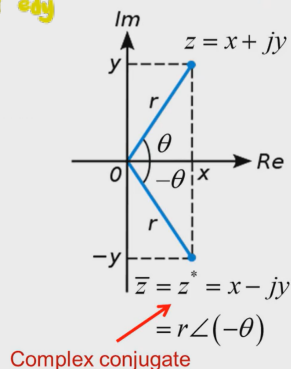
Polar form $\left\{ \begin{array}{l} = r \angle \theta \quad \text{Angle notation (phasors)} \\ = re^{j\theta} \quad \text{Exponential notation} \\ = r(\cos \theta + j \sin \theta) \quad \text{Trigonometric notation} \end{array} \right.$

Conversions:

$$j = \sqrt{-1}$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$x = r \cos \theta, \quad y = r \sin \theta$$



Addition/Subtraction: Best performed in **Cartesian form**

$$z_1 = x_1 + jy_1, \quad z_2 = x_2 + jy_2$$

$$z_1 \pm z_2 = (x_1 \pm x_2) + j(y_1 \pm y_2)$$

Multiplication/Division: Best performed in **Polar form**

$$z_1 = r_1 e^{j\theta_1}, \quad z_2 = r_2 e^{j\theta_2}$$

$$z_1 z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$$

Conjugate identities

$$z^* = re^{-j\theta}$$

$$z \cdot z^* = |z|^2 = r^2$$

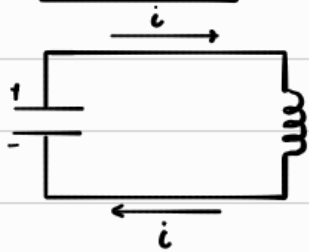
$$z^{-1} = \frac{z^*}{|z|^2}$$

Useful to calculate inverse of a complex number given in Cartesian form

Ideal LC circuit

Ideal RLC circuit is one without resistance. [a.k.a LC circuit]

Initially



Assuming at $t=0$, capacitor fully charged

1) Capacitor discharges, $V_c \downarrow \Rightarrow i \uparrow \Rightarrow V_L(t) \uparrow$

↳ Since this is a LC circuit, $V=IR$ is invalid. The following are true...

$$i_c(t) = C \frac{dV_c(t)}{dt} ; V_L(t) = L \frac{di_c(t)}{dt}$$

↳ In an inductor, Δi must be gradual, ΔV must be gradual

2) Capacitor fully discharges, $V_c = 0 \Rightarrow i_{\max} \Rightarrow V_L(t)_{\max}$

3) Even tho $V_c = 0$, i still keeps flowing because inductor keeps i in 2) flowing in the same direction.

↳ Since i continue flowing CW, means the charge continues to

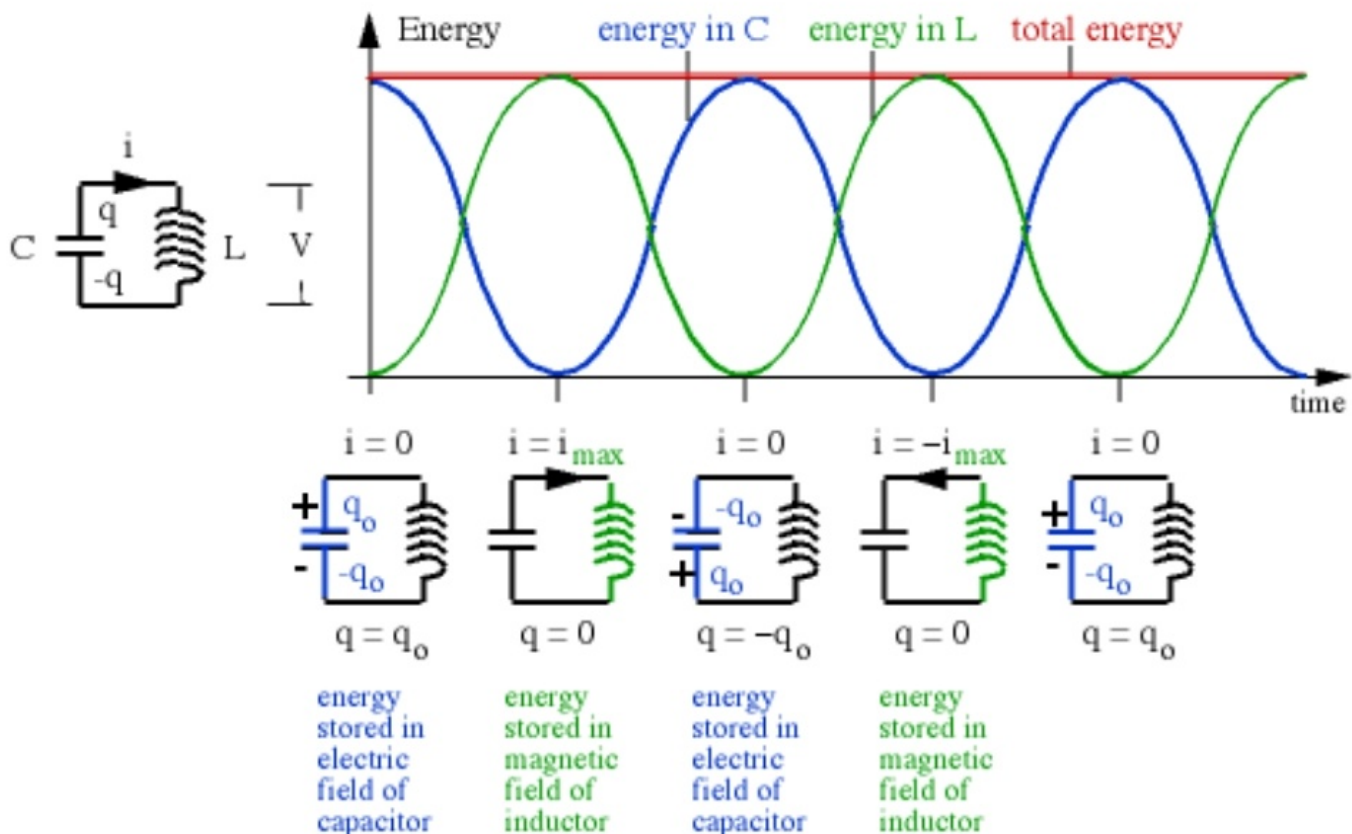
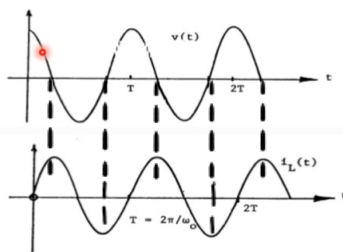
[Neutral cuz capacitor fully discharged]

leave the top neutral plate to the bottom neutral plate

This causes the top plate to be -ve, bottom plate +ve [Reversed Polarity]

4) 3) keeps happening until capacitor fully charges in reversed polarity

5) Repeat step 1 to 4 in reversed direction



1. A differential equation in x and y is an equation expressing the relationship between an **independent variable** x , a **dependent variable** y and the **derivatives** of y with respect to x .

Example 3.1

Sources = 0

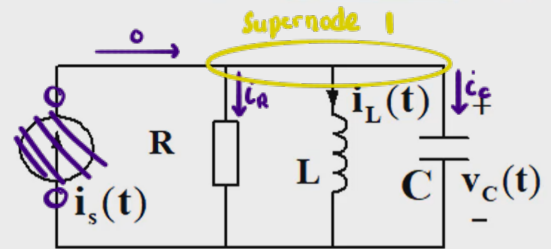
1) Same quantity

2) Coef. of highest order = 1

Only 1 Independent variable

Find the zero-input 2nd order DE in terms of $V_C(t)$

Try finding $i_L(t)$ on your own :)



Supernode 1

$$KCL : -0 + i_R(t) + i_L(t) + i_C(t) = 0$$

$$i_R(t) + i_L(t) + i_C(t) = 0$$

Basic Parallel RLC
[R, L, C all parallel]

Step 1 : Make Egn only have 1 dependent & independent variable

$$\frac{V_R(t)}{R} + \left[i_L(0) + \frac{1}{L} \int V_C(t) dt \right] + C \frac{dV_C(t)}{dt} = 0$$

Since $V_R(t) = V_L(t) = V_C(t)$; $i_L(0) = 0$

$$\frac{V_C(t)}{R} + \frac{1}{L} \int V_C(t) dt + C \frac{dV_C(t)}{dt} = 0$$

Step 2 : Make it into a 2nd order DE w
highest order coefficient = 1

Diff. both sides,

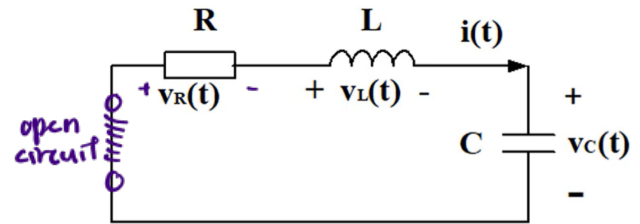
$$\frac{1}{R} \frac{dV_C(t)}{dt} + \frac{1}{L} V_C(t) + C \frac{d^2 V_C(t)}{dt^2} = 0$$

$$\frac{d^2 V_C(t)}{dt^2} + \frac{1}{RC} \frac{dV_C(t)}{dt} + \frac{1}{LC} V_C(t) = 0$$

Example 3.2

Find the zero-input 2nd order DE in terms of $i_L(t)$

Try finding $V_c(t)$ on your own :)



$$KVL : -V_R(t) - V_L(t) - V_C(t) = 0$$

$$V_R(t) + V_L(t) + V_C(t) = 0$$

Basic Series RLC
[R, L, C all series]

Step 1 : Make Eqn only have 1 dependent & independent variable

$$(i_R(t) \cdot R) + L \frac{di_L(t)}{dt} + \left[i_C(0) + \frac{1}{C} \int i_C(t) dt \right] = 0$$

Since $i_R(t) = i_C(t) = i_L(t)$; $i_C(0) = 0$

$$(i_L(t) \cdot R) + L \frac{di_L(t)}{dt} + \frac{1}{C} \int i_L(t) dt = 0$$

Step 2 : Make it into a 2nd order DE w
highest order coefficient = 1

Diff both sides ,

$$R \frac{di_L(t)}{dt} + L \frac{d^2 i_L(t)}{dt^2} + \frac{1}{C} i_L(t) = 0$$

$$\frac{d^2 i_L(t)}{dt^2} + \frac{R}{L} \frac{di_L(t)}{dt} + \frac{1}{LC} i_L(t) = 0 \quad \#$$

Standard Form of 2nd order ODE circuit

$$\textcircled{1} \frac{d^2 f(t)}{dt^2} + 2\alpha \frac{df(t)}{dt} + \omega_0^2 f(t) = 0$$

Damping constant

1) Parallel RLC = $\frac{1}{2RC}$

2) Series RLC = $\frac{R}{2L}$

[Decay Tendency]

Undamped Natural Freq.

$\frac{1}{\sqrt{LC}}$ for parallel & series RLC

[Oscillation Tendency]

$$\textcircled{2} s^2 + 2\alpha s + \omega_0^2 = 0 \quad ; \quad s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

[Characteristic Eqn]

$$\textcircled{3} \frac{d^2 f(t)}{dt^2} + 2\omega_0 \zeta \frac{df(t)}{dt} + \omega_0^2 f(t) = 0$$

Damping ratio

$\zeta = \alpha/\omega_0 = \frac{1/R}{2C}$

[Normalized damping]

$$\textcircled{4} \frac{d^2 f(t)}{dt^2} + \frac{\omega_0}{Q} \frac{df(t)}{dt} + \omega_0^2 f(t) = 0$$

Quality factor

Parallel = $\omega_0 RC$

Series = $\omega L/R$

[Energy-preserving tendency]

Take Note:

An ideal LC circuit oscillates forever.

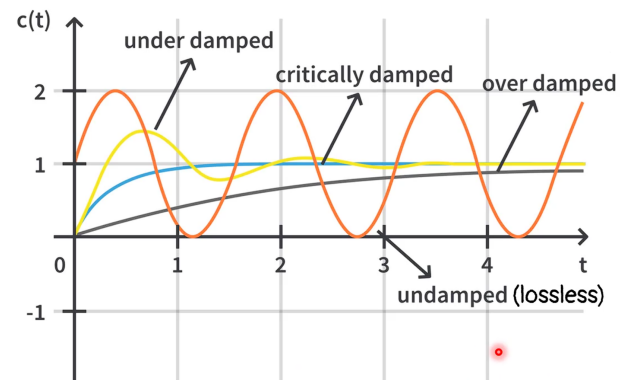
1) When $i \uparrow$, L creates EMF in the opposite direc.

to oppose the incre. in current

2) when $i \downarrow$ to zero, L creates EMF in the opposite direc.

to oppose the decre. in current

3) This charges back C to the opposite polarity and the cycle repeats



Parallel $\alpha = \frac{1}{2RC}$
Series $\alpha = R/2L$

	OVERDAMPED	CRITICALLY DAMPED	UNDERDAMPED	LOSSLESS
Roots	Both real & negative	Equal, real & negative	Complex conjugate	Imaginary
s_1, s_2	$-\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	$-\alpha, -\alpha$	$-\alpha \pm j\omega_d$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	$\pm j\omega_0$
Damping const α	$\alpha > \omega_0$	$\alpha = \omega_0$	$0 < \alpha < \omega_0$	$\alpha = 0$
Quality factor $Q \propto \frac{1}{\zeta}$	$0 < Q < 0.5$	$Q = 0.5$	$Q > 0.5$	$Q \rightarrow \infty$
Damping ratio $\zeta \propto \alpha/\omega_0$	$\zeta > 1.0$	$\zeta = 1.0$	$0 < \zeta < 1.0$	$\zeta = 0$
homogenous response (series & parallel)	$k_1 e^{s_1 t} + k_2 e^{s_2 t}$	$(k_1 + k_2 t) e^{-\alpha t}$	$k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)$	

$$I(t) = \frac{1}{L} \int_0^t V_L(t) dt + V_L(0)$$

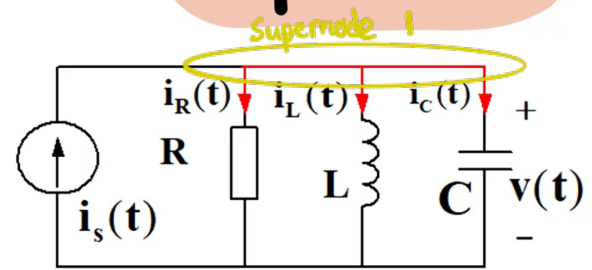
$$V_C(t) = \frac{1}{C} \int_{t_0}^t i_C(t) dt + i_C(0)$$

Example 3.3

Find the 2nd-order Zero-state response for $i_L(t)$.

$$i_s(t) = u(t)$$

$$i_L(0^-) = 0, v_C(0^-) = 0$$



Supernode 1

$$KCL : i_s(t) - i_R(t) - i_L(t) - i_C(t) = 0$$

$$i_s(t) = i_R(t) + i_L(t) + i_C(t)$$

Step 1 : Make Eqn only have 1 dependent & independent variable

$$i_s(t) = \frac{V_R(t)}{R} + i_L(t) + C \frac{dV_C(t)}{dt}$$

Since $V_R(t) = V_L(t) = V_C(t)$,

$$i_s(t) = \frac{V_L(t)}{R} + i_L(t) + C \frac{dV_L(t)}{dt}$$

$$= \frac{1}{R} \frac{d i_L(t)}{dt} + i_L(t) + LC \frac{d^2 i_L(t)}{dt^2}$$

Step 2 : Make it into a 2nd order DE w/ highest order coefficient = 1

$$\frac{i_s(t)}{LC} = \frac{d^2 i_L(t)}{dt^2} + \frac{1}{RC} \frac{d i_L(t)}{dt} + \frac{1}{LC} i_L(t)$$

Converting to characteristic eqn,

$$0 = s^2 + \frac{1}{RC} s + \frac{1}{LC}$$

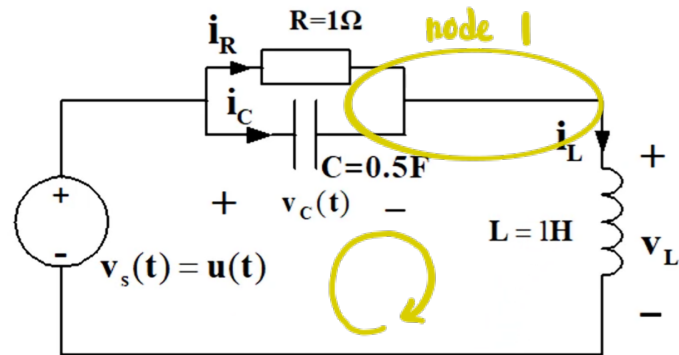
If we are given R & C values, we just have to solve the quadratic to get S.

Example 3.4

a) Find:

$i_R(0+)$, $i_L(0+)$, $V_C(0+)$

$i_R(\infty)$, $i_L(\infty)$, $V_C(\infty)$



b) Find the characteristic roots of the circuit.

Is the circuit overdamped, critically damped, or underdamped?

c) Derive the full expression for $v_C(t)$. Show step-by-step procedure.

d) Sketch the waveform of the step response $v_C(t)$. Indicate the salient features of the waveform.

$$b) \quad KCL : i_L(t) = i_R(t) + i_C(t) = 0 \quad \text{--- ①}$$

$$KVL : V_s(t) - V_C(t) - V_L(t) = 0 \quad \text{--- ②}$$

Step 1: Make Eqn only have 1 dependent & independent variable

$$i_L(t) = \frac{V_C(t)}{R} + C \frac{dV_C(t)}{dt}, \text{ using ①, } [V_R(t) = V_C(t)] \quad \text{--- ③}$$

Step 2: Make it into a 2nd order DE w/ highest order coefficient = 1

$$\begin{aligned} V_s(t) &= V_C(t) + V_L(t), \text{ using ②} \\ &= V_C(t) + L \frac{di_L(t)}{dt} \\ &= V_C(t) + L \frac{d}{dt} \left[\frac{V_C(t)}{R} + C \frac{dV_C(t)}{dt} \right], \text{ [using ③]} \\ &= V_C(t) + \frac{L}{R} \frac{dV_C(t)}{dt} + LC \frac{d^2V_C(t)}{dt^2} \end{aligned}$$

$$\frac{V_s(t)}{LC} = \frac{d^2V_C(t)}{dt^2} + \frac{1}{RC} \frac{dV_C(t)}{dt} + \frac{1}{LC} V_C(t)$$

Converting to characteristic eqn,

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

$$s^2 + 2s + 2 = 0, \text{ [using } R=1, C=0.5, L=1]$$

$$s = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm j$$

$$\therefore \text{Under damping, } \alpha = 1, \omega_d = 1 \text{ [} s = -\alpha \pm j\omega_d \text{]}$$

c) Since underdamped, homogenous solution = $k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)$

$$\therefore V_C(t) = \underbrace{[k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)]}_{\text{homogenous}} + \underbrace{V_C(\infty)}_{\text{particular}}$$

$$= k_1 e^{-t} \cos(t) + k_2 e^{-t} \sin(t) + 1$$

$$V_C(0) = k_1(1)(1) + k_2(1)(0) + 1$$

$$0 = k_1 + 1$$

$$k_1 = -1 \quad \#$$

Rmb $\frac{dV_C(0^+)}{dt} \neq \frac{d}{dt}(0)$
since the gradient at $t=0$ might not be 0

$$V_C(\infty) = k_1 e^{-\infty} \cos(\infty) + k_2 e^{-\infty} \sin(\infty) + 1$$

$$1 = 1$$

[Useless]

$$\frac{dV_C(t)}{dt} = k_1 [-e^{-t} \sin(t) - e^{-t} \cos(t)] + k_2 [e^{-t} \cos(t) - e^{-t} \sin(t)] + 0$$

$$\frac{dV_C(0^+)}{dt} = -1 [-1(0) - 1(1)] + k_2 [1(1) - 1(0)] \quad , \text{ using } k_1 = -1$$

$$\frac{dV_C(0^+)}{dt} = 1 + k_2 \quad \text{--- (4)}$$

steady state
capacitor = open circuit

$$i_C(0^+) = C \frac{dV_C(0^+)}{dt}$$

$$0 = C \frac{dV_C(0^+)}{dt}$$

$$\therefore \frac{dV_C(0^+)}{dt} = 0 \quad \text{since } C \neq 0$$

$\therefore$ Using eqn (4),

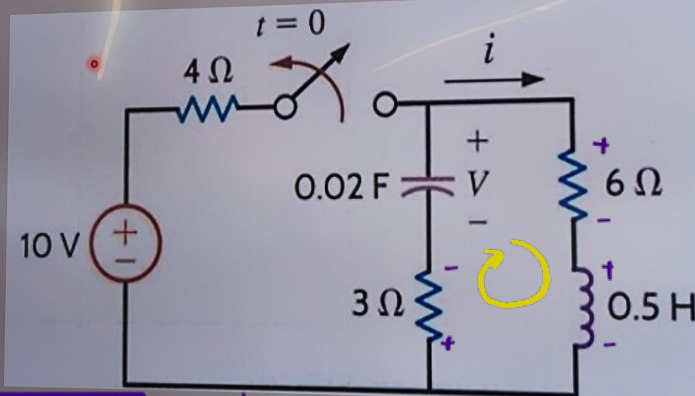
$$0 = 1 + k_2$$

$$k_2 = -1 \quad \#$$

$$\therefore V_C(t) = -e^{-t} \cos(t) - e^{-t} \sin(t) + 1 \quad \text{V} \quad \#$$

Example 1

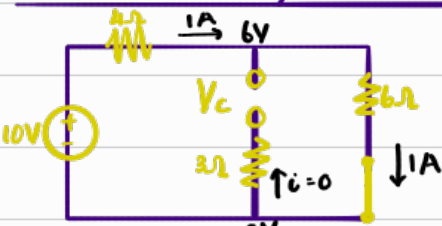
Find $i(t)$ in the circuit for $t > 0$.



At $t > 0$, an RLC circuit might oscillate according to the damping that we have yet to find. Thus, since we do not know the polarity of C and L, we just use the basic KVL rule that is $\sum V = 0$ around a loop w/out setting any polarity.

Steady state conditions

Initial conditions, $t < 0$



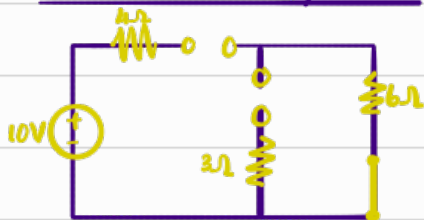
$$i(0^-) = \frac{10}{4+6} = 1.0 \text{ A}$$

$$i(0^+) = i(0^-) = 1.0 \text{ A}$$

caz of inductor

$$V_C(0) = 10\text{V} - V_{4\Omega} = 10 - 1(4) = 6\text{V}$$

Final conditions, $t \rightarrow \infty$



$$i(\infty) = 0, \text{ capacitor loses charge}$$

Transient Response, $t > 0$

$$\text{KVL: } +V_{6\Omega} + V_L + V_{3\Omega} + V_C = 0$$

Step 1: Make Eqn only have 1 dependent & independent variable

$$+i(6) + V_L + i(3) + V_C = 0$$

$$q i_L(t) + L \frac{di_L(t)}{dt} + \frac{1}{C} \int i_L(t) dt + V_C(0) = 0, [i = i_L = i_C]$$

$$q i_L(t) + L \frac{di_L(t)}{dt} + \frac{1}{C} \int i_L(t) dt + V_C(0) = 0, [i = i_L = i_C]$$

$$q \frac{di_L(t)}{dt} + L \frac{d^2 i_L(t)}{dt^2} + \frac{1}{C} i_L(t) + V_C(0) = 0$$

$$\frac{d^2 i_L(t)}{dt^2} + \frac{q}{L} \frac{di_L(t)}{dt} + \frac{1}{LC} i_L(t) + 0 = 0$$

Converting to characteristic eqn,

$$s^2 + \frac{q}{L}s + \frac{1}{LC} = 0$$

$$s^2 + 18s + 100 = 0, [C = 0.02\text{F}, L = 0.5\text{H}]$$

$$\therefore s = -9 \pm \sqrt{19} j, \text{ underdamping}$$

$$\alpha = 9, \omega_d = \sqrt{19}$$

c) Since underdamped, homogenous solution =

$$\therefore i_L(t) = [k_1 e^{-9t} \cos(\sqrt{19}t) + k_2 e^{-9t} \sin(\sqrt{19}t)] + i_L(\infty)$$

$$= k_1 e^{-9t} \cos(\sqrt{19}t) + k_2 e^{-9t} \sin(\sqrt{19}t) + 0 \quad \text{--- ①}$$

$$i_L(0) = k_1(1)(1) + k_2(1)(0) + 0 \quad \left| \quad i(\infty) = 0 + 0 + 0 \right.$$

$$1 = k_1$$

$$0 = 0 \text{ [useless]}$$

NEXT PG

Diff ①,

$$\frac{di_L(t)}{dt} = k_1 [-\sqrt{19} e^{-9t} \sin(\sqrt{19} t) - 9 e^{-9t} \cos(\sqrt{19} t)] + k_2 [\sqrt{19} e^{-9t} \cos(\sqrt{19} t) - 9 e^{-9t} \sin(\sqrt{19} t)]$$

$$\frac{di_L(0^+)}{dt} = 1 [0 - 9(1)(1)] + k_2 [\sqrt{19}(1)(1) - 0], \quad [\text{using } k_1 = 1]$$

$$\frac{di_L(0^+)}{dt} = -9 + k_2(\sqrt{19}) \quad \text{--- ②}$$

$$V_L(0^+) = L \frac{di_L(0^+)}{dt} \quad \text{--- ③}$$

At $t = 0^+$, i is definitely still flowing in the same direction as $t = 0^-$.
So we can just consistently set
+ terminal $\Rightarrow$ current in
- terminal $\Rightarrow$ current out
OR Vice Versa

At $t = 0^+$ $[i(0^+) = 1]$

$$\text{KVL: } -V_{3\Omega} + V_C(0) - V_{6\Omega} - V_L(0) = 0$$

$$-1(3) + 6 - 1(6) - V_L(0) = 0$$

$$V_L(0) = -3 \text{ V} \quad \text{--- ④}$$

$$\text{④} \rightarrow \text{③}$$

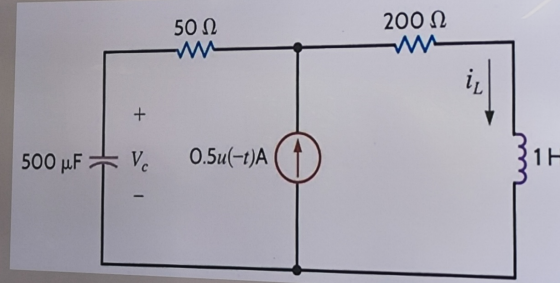
$$-3 = 0.5 [-9 + k_2 \sqrt{19}], \quad [\text{using ②}]$$

$$k_2 = \frac{3}{\sqrt{19}}$$

$$\therefore i_L(t) = e^{-9t} \cos(\sqrt{19} t) + \frac{3}{\sqrt{19}} e^{-9t} \sin(\sqrt{19} t) \quad \#$$

Question 1

For the circuit shown below, find $i_L(t)$ for $t > 0$.



$$u(t) \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

$$0.5u(t) \begin{cases} 0 & \text{for } t < 0 \\ 0.5 & \text{for } t \geq 0 \end{cases}$$

$$0.5u(-t) \begin{cases} 0 & \text{for } t > 0 \\ 0.5 & \text{for } t \leq 0 \end{cases}$$

$$t < 0$$

$$i_L(0) = 0.5 \text{ A}$$

$$t \rightarrow \infty$$

$$i_L(\infty) = 0 \text{ A}$$

$$t > 0$$

$$V_C(t) + V_R(t) + V_L(t) = 0$$

$$\left[\frac{1}{C} \int i_C(t) dt + V_C(0) \right] + (i_L(t) \cdot R) + L \frac{di_L(t)}{dt} = 0$$

Diff both sides,

$$\frac{1}{C} i_L(t) + 0 + R \frac{di_L(t)}{dt} + L \frac{d^2 i_L(t)}{dt^2} = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$s^2 + 250s + \frac{1}{500 \mu\text{F}} = 0, \text{ [using } L = 1 \text{ H, } C = 500 \mu\text{F}]$$

$$\therefore s_1 = -8.3, s_2 = -242 \text{ overdamping.}$$

$$\begin{aligned} \therefore i_L(t) &= k_1 e^{s_1 t} + k_2 e^{s_2 t} + i_L(\infty) \\ &= k_1 e^{-8.3t} + k_2 e^{-242t} + 0 \end{aligned}$$

$$i_L(0) = k_1 + k_2$$

$$k_1 + k_2 = 0.5$$

$$\frac{di_L(t)}{dt} = [-8.3 k_1 e^{-8.3t} - 242 k_2 e^{-242t}]$$